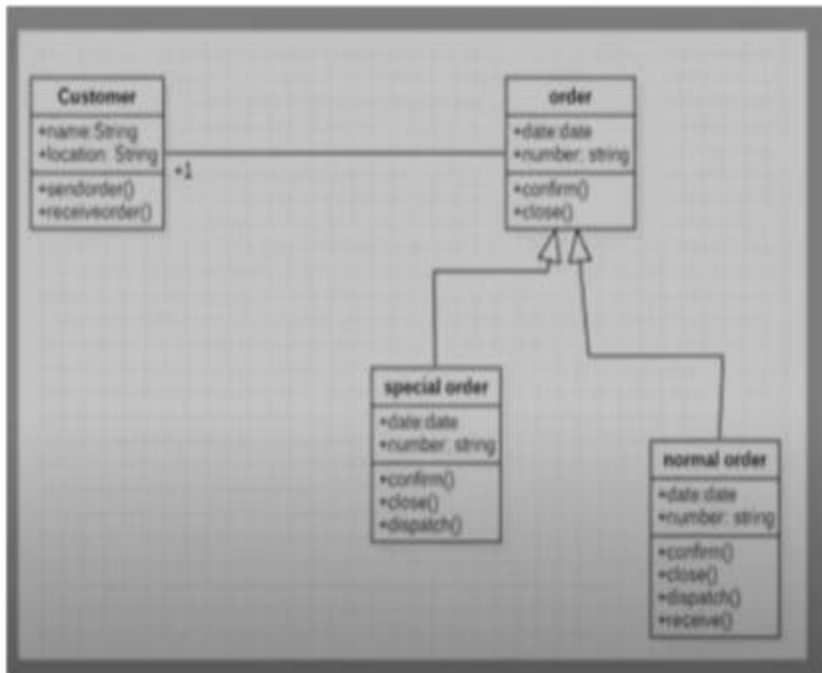


Place Order

Question:

4. Write a Java Program to below class diagram:



Step wise execution

1. The program starts execution from the `main()` method.
2. A **Customer** object `c` is created with:
 `name = "susmitha"`
 `location = "Anits"`
3. The **Customer** constructor initializes the name and location.
4. `c.sendorder()` is called.
5. The following message is displayed:
 susmitha from Anits is dropped an order, no cap

6. `c.receiveorder()` is called.

7. The following message is displayed:

order delivered successfully to susmitha

8. A `NormalOrder` object `n` is created with:

`date = "12-09-2044"`

`no = "P1234O12"`

9. The `NormalOrder` constructor calls the parent

`Order` constructor using `super()`.

10. The `Order` constructor initializes `date` and `no`.

11. `n.confirm()` is called.

12. The overridden `confirm()` method of `NormalOrder`

displays the normal order confirmation.

13. `n.dispatch()` is called.

14. The standard delivery message is displayed.

15. `n.receive()` is called.

16. The message that the normal order is received

is displayed.

17. `n.close()` is called.

18. The overridden `close()` method of `NormalOrder`

displays the order closing message.

19. A `SpecialOrder` object `s` is created with:

`date = "17-3-3098"`

`no = "Q23234"`

20. The `SpecialOrder` constructor calls the parent

Order constructor using super().

21. The Order constructor initializes date and no.
22. s.confirm() is called.
23. The overridden confirm() method of SpecialOrder displays the special order confirmation.
24. s.dspatch() is called.
25. The express delivery message is displayed.
26. s.close() is called.
27. The overridden close() method of SpecialOrder displays the order closing message.
28. After all operations are completed, the program terminates.

Algorithm

1. Start the program.
2. Define the Customer class with name and location.
3. Create a Customer constructor to initialize name and location.
4. Define sendorder() to display the customer order message.
5. define receiveorder() to display the successful delivery message.
6. Define the Order class with date and order number.
7. Create an Order constructor to initialize date

and order number.

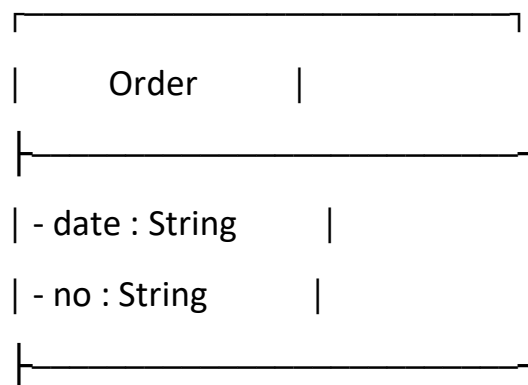
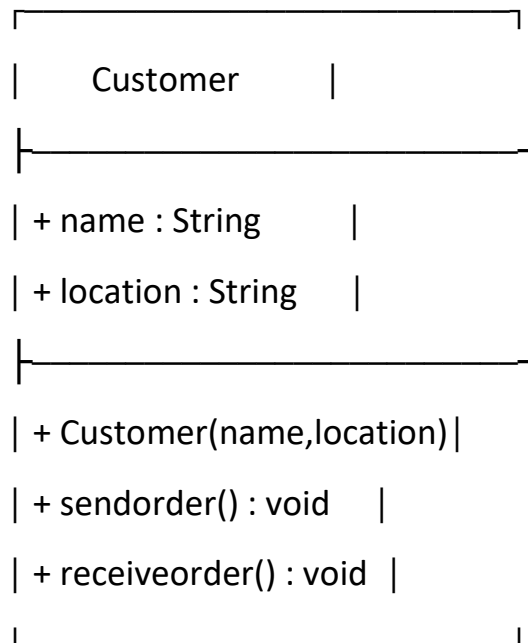
8. Define confirm() to display the order confirmation.
9. Define close() to display the order closing message.
10. Define SpecialOrder as a subclass of Order.
11. Override confirm() and close() methods in SpecialOrder.
12. Define dispatch() in SpecialOrder to display express delivery information.
13. Define NormalOrder as a subclass of Order.
14. Override confirm() and close() methods in NormalOrder.
15. Define dispatch() in NormalOrder to display standard delivery information.
16. Define receive() in NormalOrder to display order received information.
17. Start the main() method.
18. Create a Customer object with name "susmitha" and location "Anits".
19. Call sendorder() and receiveorder().
20. Create a NormalOrder object with its date and order number.
21. Call confirm(), dispatch(), receive(), and close() for the NormalOrder.
22. Create a SpecialOrder object with its date and

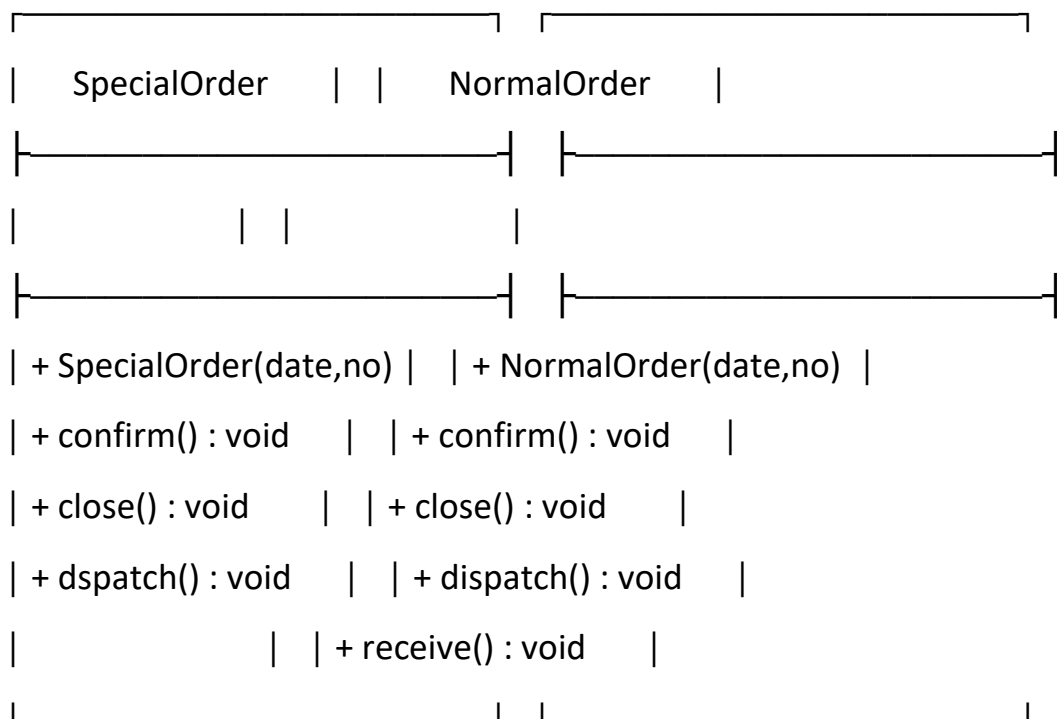
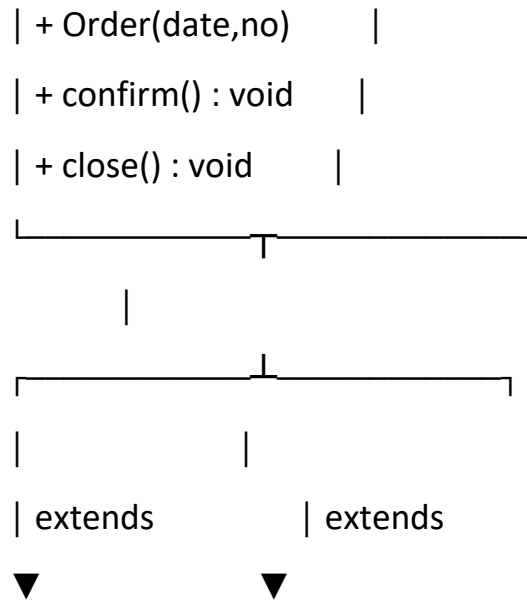
order number.

23. Call `confirm()`, `dspatch()`, and `close()` for the `SpecialOrder`.

24. Stop the program.

UML diagram





Code

```

package oops;

class Customer{

```

```

    public String name;
    public String location;

    public Customer(String name,String location){
        this.name=name;
        this.location=location;
    }

    void sendorder(){
        System.out.println(name + " from " + location + " is
dropped an order, no cap");;
    }
    void receiveorder(){
        System.out.println("order delivered successfully to " +
name + "\n");
    }
}

class Order {
    String date;
    String no;
    public Order(String date,String no){
        this.date=date;
        this.no=no;
    }

    void confirm(){
        System.out.println("order" + no + " is placed on " +
date);
    }
    void close(){
        System.out.println("order# " + no + " closed");
    }
}

class SpecialOrder extends Order {

```

```

    public SpecialOrder(String date, String no) {
        super(date, no);
    }

    @Override
    void confirm(){
        System.out.println("special order " + no + " is placed
on " + date);
    }

    @Override
    void close(){
        System.out.println("special order# " + no + "
closed\n");
    }

    void dspatch(){
        System.out.println("special order # " + no + " is
dispatched via express delivery");
    }
}

class NormalOrder extends Order{
    public NormalOrder(String date, String no) {
        super(date, no);
    }

    @Override
    void confirm(){
        System.out.println("normalorder " + no + " is placed on
" + date);
    }

    @Override
    void close(){

```



```

        System.out.println("normal order # " + no + "
closed\n");
    }

    void dispatch(){
        System.out.println("normal order #" + no + " is
dispatched via standard delivery");
    }

    void receive(){
        System.out.println("normal order # " + no + " is
received");
    }
}

public class PlaceOrder {
    Customer c = new Customer("susmitha", "Anits");

    public static void main(String[] args) {
        Customer c = new Customer("susmitha", "Anits");
        c.sendorder();
        c.receiveorder();

        NormalOrder n=new NormalOrder("12-09-2044","P1234012");
        n.confirm();
        n.dispatch();
        n.receive();
        n.close();

        SpecialOrder s=new SpecialOrder("17-3-3098","Q23234");
        s.confirm();
        s.dspatch();
        s.close();
    }
}

```

```
}
```

Output screen shot

```
PS C:\Users\Admin\OneDrive\Documents\basicjava>
PS C:\Users\Admin\OneDrive\Documents\basicjava> c;; cd 'C:\Users\Admin\OneDrive\Documents\basicjava'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+showCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Admin\OneDrive\Documents\basicjava\bin' 'oops.PlaceOrder'
susmitha from Anits is dropped an order, no cap
order delivered successfully to susmitha

normalorder P1234012 is placed on 12-09-2044
normal order #P1234012 is dispatched via standard delivery
normal order # P1234012 is received
normal order # P1234012 closed

special order Q23234 is placed on 17-3-3098
special order # Q23234 is dispatched via express delivery
special order# Q23234 closed

PS C:\Users\Admin\OneDrive\Documents\basicjava> |
```